Introduction

Jose Manuel Paz y Miño

Universidad de Chile FEN

2026

Introduction

- Welcome to Empirical Industrial Organization.
- This course mixes IO + econometrics + tons of coding.
- Even if you are not particularly fond of IO, it will teach you a framework to analyze strategic interactions among agents (more about this later).

Contents

- We can divide the course into several parts.
  * Preliminaries: Numerical methods, GMM-IV.
  * Demand Estimation (homogeneous and heterogeneous products).
  * Supply Estimation: Conduct, Entry/Exit of Products, Repeated Games.
  * Other topics: Markups/Markdowns, Boycott, Non-IO Applications.

A Tree in a Forest

- We are focusing on (a subset of) static problems.
- What are we not covering?: Consumer Search, Two-Side markets, Bargaining and matching, Entry/Exit, Dynamic Decision Models, Dynamic Games.
- These all could be covered in a second Empirical IO course.
- But don't worry, our course should give you a strong foundation for these other topics.
- **We can create a second course (reading group) if there is enough demand.**

Types of Empirical IO papers

- 3 types
  1. Some complex new estimation method.
  2. A careful analysis of public policy that needs to consider strategic interactions among agents.
  3. Point 1 + 2.

- This course should help you with point 2[1]. I can help with point 1 (or 3) once I get tenure.

[1] which is way beyond what is needed for a ME/MAE thesis and exactly correct for a Ph.D. one.

Where to find papers?

- A Ranking
  1. Top 5+1: AER, RESTUD, ECMA, JPE, QJE, RES.
  2. Top Fields: RAND, AEJ:MICRO, AEJ:AE, JPE:MICRO, JPubE, JIE, JLE, MS, MktS, JUE
  3. 2nd Tier IO: IJIO, JIndE, JELO, JEMS

- Empirical IO papers are difficult as they demand that you master lots of topics but when published, they are usually at high ranked journals.

Relation to public policy

- We will see some pretty cool models and discuss technical aspects, but (for me) the strongest point of these is that we can evaluate different policies. These can have impact on
  * Changes in consumer surplus (higher prices, lower varieties). Applies to any type of market (education, health, environmental, political, etc).
  * Redistribution.
  * Passthrough of shocks.
  * Tax policy.
  * Antitrust: merger policy, damage estimation, etc.
  * etc.

Things I will assume

- Comfortable reading any chapter in [Tirole, 1988].
- **Willing to learn how to code in Python. This is not a coding course. I'll provide sample code, but most of the learning will be done offline by you. Tons of online resources.**
- Willing to read selected papers **and be ready to discuss them in class**.

Some cool thesis topics I'm advising / have advised

- Impact of climate change in the international coffee market.
- Analysis of merger remedies in the Fiat/Peugeot merger in Chile.
- Testing the *deposit channel* of monetary policy in the US Bank market.
- Demand-side effect of trade liberalization during NAFTA in the US Beer Market.
- Benefits and incidence of military intervention in the cellulose pulp Chilean export market.
- Value of Tax-Coordination in the Gran Buenos Aires Fuel Market.
- Luxury Car taxes in the Chilean Market.
- Advertisement equilibrium effects of no-reelection of mayors in

Other Interesting Markets (Chilean Data)

- Forestry.
- Fishing.
- Housing.
- Airlines.
- Mining.
- Others?

More Resources

- Check replication codes at AER, QE, AEJ:Micro, AEJ:AE, AEJ:EP, RESTUD, ECMA or RES.
- Chris Conlon's graduate IO (and econometrics) classes and videos.
- Victor Aguirregabiria's unpublished empirical IO book.
- Dynamic Structural Econometric Youtube channel and Github site.
- Shoshana Vasserman's Industrial Organization Job Market Candidates website.
- ...

The Elephant in the room

About the use of ChatGPT/Claude. Some ideas.

- Greatly improves performance at the cost of learning less.
  * I suffered coding all this by hand. I can tell when these spit some non-sense, can you?
- Useful for students and professors.
- Challenge to make homeworks such that questions go beyond what ChatGPT/Claude usually can answer.
- **It makes mistakes.**
- An interview at Amazon usually involves writing the BLP algorithm by hand on a board. Can you ChatGPT'd this?

Grading

4 problems sets (20%) + Quizzes (at the beginning of class on selected papers)

Do we really know BLP? (Exhibit 1)

econjobrumors.com/topic/structural-modeling-track-amazon

Economist
86e7

I went through this a while ago.

The questions were Amazon business related questions where the answer was “run a BLP style model.”

It was relevant stuff from their business, like “what would we want to know if we offered <product>?”

Then you get into the BLP details. Not implementation like the guy above said but...
- why that model
- what data do you need
- how does the model work (roughly)
- what do the results mean
- how would you use the model you estimated to answer X question related to the one we posed

Also in the structural track they will ask you about dynamics at some point

2 YEARS AGO # QUOTE 9 GOOD 0 NO GIOD !

Do we really know BLP? (Exhibit 2)

econjobrumors.com/topic/how-long-did-it-take-you-to-fully-understand-blp

Economist
5782

Took me good two days to read through Nevo's guide, and a full day to code it up.

And still feel like i need to read the paper over a couple more times to fully absorb it.

I am a first year student who wants to do IO, and this seems a bit slow... :(

4 YEARS AGO # QUOTE 3 GOOD 0 NO GIOD !

Economist
0d43

I have a top 5 and do IO and I'm not sure I understand it really.

4 YEARS AGO # QUOTE 13 GOOD 0 NO GIOD !

Do we really know BLP? (Exhibit 3)

econjobrumors.com/topic/can-someone-please-explain-blp-to-me/page/2

Economist
4cec

By experience, it may take around two(!) months to grasp the technique (at level when you are able to rewrite it in different programming language, understanding what you are doing at each step).

Everyone should definitely study the BLP model to see the interplay of several useful concepts (logit, contraction mapping - Newton-Rapson and gradient descent, instrumental variables, GMM, intagration, multidimentional distribution, numerical solution of system of non-linear equations, jacobian (gradient), just to name a few).
You can hardly find a better example of so useful combination of methods.
Apart from math and economics, you may learn how to structure and solve optimization problems in Matlab (which is also pritty useful).
If you are not good at any of that things, issues with understanding are guarantied. But be ready for that and do not give up. The result is going to be encouraging (you will definitely rise your self-confidence).

If you do not understand the pure analytical writings (sometimes written with errors, or even misleading in inportant places), use coding (i.e. Matlab version https://eml.berkeley.edu/~bhhall/e220c/rc_dc_code.htm).

Nevo + Rasmusen + some later sources mentioned in Rasmusen are all useful. (The original paper is not needed at all.)

6 YEARS AGO # QUOTE 8 GOOD 0 NO GIOD !

Next Class

- Read [Haile, 2024] and be ready to discuss it in class.
- We start the introduction of numerical methods.

Haile, P. (2024).
Models, measurement, and the language of empirical economics.

Technical report.

Tirole, J. (1988).
The theory of industrial organization.
MIT press.